ANDROID SPATIAL COMPUTING

Building Android XR

Developing Spatial Experiences with Jetpack XR SDK and Android Studio

ECOSYSTEM OVERVIEW

The Android XR OS Platform

A unified open operating system built by Google, Samsung, and Qualcomm—bringing standard Android apps and immersive 3D experiences into headset spatial computing.

Home Space vs Full Space Modes

Home Space Mode

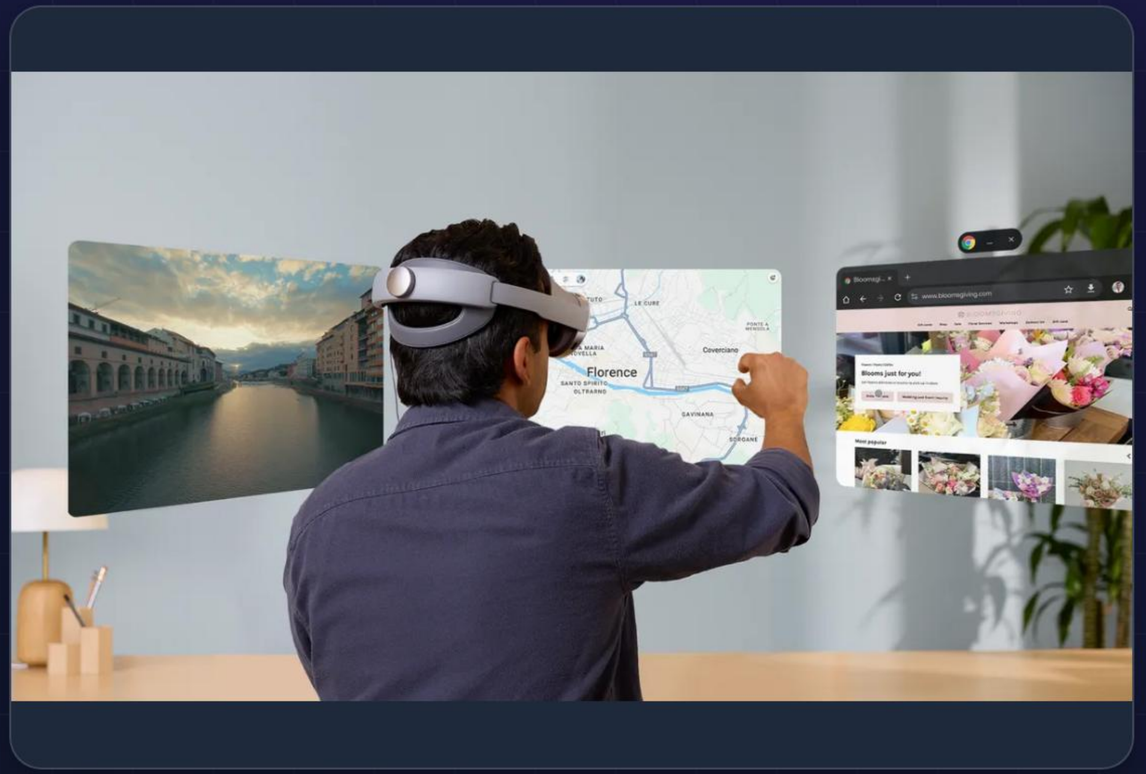
Multiple 2D and 3D application windows run side-by-side in spatial environment. Supports familiar Android multitasking with resizable spatial panels, orbiters, and system windowing controls.

Full Space Mode

The app unlocks boundless 3D spatial freedom. Enables immersive environment overrides, direct 3D mesh rendering, hand tracking interactions, and spatial anchors anchored to real-world physical space.

Android Studio XR Tooling

- **Basic Headset Activity Template:** Native project starter preconfigured with Jetpack XR dependencies and spatial manifest flags.
- **Android XR Emulator:** Built-in virtual headset device simulating head-pose tracking, controller input, and passthrough.
- **Compose Layout Previews:** Real-time rendering of spatial UI panels, orbiters, and depth hierarchy directly inside the IDE.
- **Jetpack SceneCore Debugger:** Live inspection of 3D entities, spatial anchors, lighting, and materials in real time.



Jetpack XR Core SDK Stack



Jetpack Compose for XR

Declarative spatial UI building blocks including SpatialPanels, Orbiters, Subspace layouts, and Material Design spatial components for Kotlin developers.



Jetpack SceneCore

High-performance entity component framework for loading glTF 3D models, customizing PBR materials, controlling light sources, and triggering spatial animations.



ARCore for XR

Low-level spatial perception engine powering real-world depth estimation, 6DoF head tracking, gesture/hand tracking, and spatial anchors.

100% Android App Compatibility

100%

Android Ecosystem Reach

Zero Codebase Rewrite Required

Standard Android applications run automatically in Android XR Home Space as high-resolution spatial panels. Developers can immediately publish existing Google Play apps and progressively spatialize UI elements using Jetpack Compose for XR without rebuilding from scratch.

UI PATTERNS

Spatial UX Design & Containment

Uncontain and Elevate

Designing for headsets requires shifting from rigid flat screens to spatial depth. Remove heavy background containers to let content breathe in 3D space.
Utilize orbiters for contextual controls and establish clear spatial visual anchors so users remain grounded and comfortable within mixed reality environments.



XR Framework Comparison Table

FEATURE / CAPABILITY	JETPACK XR SDK (NATIVE)	UNITY 6 + OPENXR	WEBXR / PWA
Primary Language	Kotlin / Jetpack Compose	C# / ShaderLab	JavaScript / WebGL
Gemini AI Integration	Deep Native System Integration	Plugin / API level	REST APIs
Standard 2D App Spatialization	Native (Zero extra overhead)	Requires custom canvas wrapper	Browser container window
Ideal Use Case	Utility, Productivity, Enterprise	High-end 3D Games, VR Simulations	Lightweight Web Experiences

Latency and Frame Rate Benchmarks



Native Jetpack XR builds achieve maximum motion-to-photon responsiveness and maintain stable 90 Hz display refresh rates on modern Snapdragon XR hardware.

XR App Development Timeline

Phase 1

Setup Android Studio & create project via Basic Headset Activity template.

Phase 2

Spatialize UI with Compose XR, Orbiters, and Jetpack SceneCore entities.

Phase 3

Test and iterate spatial ergonomics using the Android Studio XR Emulator.

Phase 4

Deploy & test on hardware like Samsung Galaxy XR headset.

"Spatial computing is not about replacing the screen—
it is about removing the boundary between software
and the physical world."

— ANDROID XR DEVELOPER VISION

Build Future Experiences Today

Get started with the Jetpack XR SDK and Android Studio

developer.android.com/xr | github.com/android/xr-samples

Image Sources



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Source: www.androidcentral.com



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